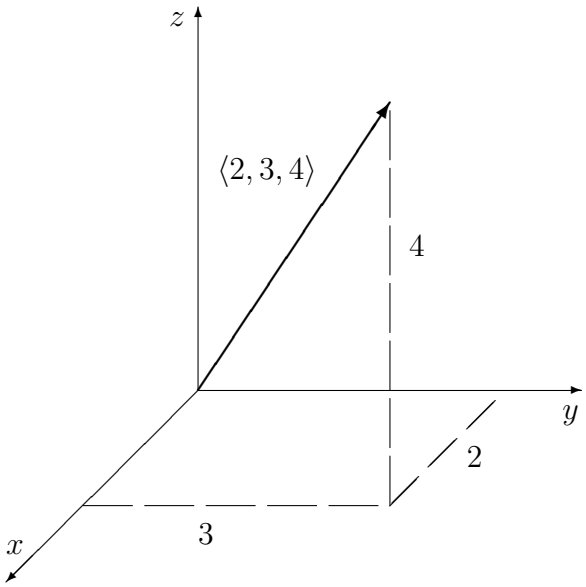
Vectors in $\mathbb{R}^3$

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Here, I will describe some of the basic mathematical operations on vectors in 3-dimensional space. A vector in 3-dimensional space is given by 3 coordinates: $\langle x, y, z\rangle$ where each of those coordinates represents a distance/difference in that direction. We draw a vector by choosing a starting point, and then drawing an arrow from there to its endpoint, which is found by moving those three distances from the starting point. For example: If we draw the vector $\langle 2, 3, 4\rangle$, starting at the origin:



I will use the following notations in the formulas and examples that follow:

- $\vec{a} = \langle 4, -1, 2\rangle$
- $\vec{b} = \langle -2, 3, 5\rangle$
- $\vec{u} = \langle x_u, y_u, z_u\rangle$
- $\vec{v} = \langle x_v, y_v, z_v\rangle$
- m is a Real Number

Properties and operations with vectors in $\mathbb{R}^3$:

Length: The length of a vector in $\mathbb{R}^3$ is computed by using the 3-dimensional version of the Pythagorean Theorem, and can be denoted using either the single-bar absolute value symbol or the double-bar norm symbol (people use the double-bar symbol when they want to show off how much abstract analysis they've studied):

$$|\vec{u}| = \|\vec{u}\| = \sqrt{{x_u}^2 + {y_u}^2 + {z_u}^2}$$

EXAMPLE: $|\vec{a}| = \sqrt{4^2 + (-1)^2 + 2^2} = \sqrt{16 + 1 + 4} = \sqrt{21}$

Addition: Two vectors in $\mathbb{R}^3$ can be added simply by adding each coordinate:

$$\vec{u} + \vec{v} = \langle x_u + x_v, y_u + y_v, z_u + z_v \rangle$$

EXAMPLE: $\vec{a} + \vec{b} = \langle 4, -1, 2 \rangle + \langle -2, 3, 5 \rangle = \langle 2, 2, 7 \rangle$

Subtraction: Subtraction is similar to addition:

EXAMPLE: $\vec{a} - \vec{b} = \langle 4, -1, 2 \rangle - \langle -2, 3, 5 \rangle = \langle 6, -4, -3 \rangle$

Multiplication: Multiplication can be carried out in three different ways:

1. Scalar Multiplication: You can multiply a vector by a real number, by multiplying each coordinate by that real number, the result is a longer/shorter vector in $\mathbb{R}^3$ which points in the same or opposite direction:

   $$m\vec{u} = m \langle x_u, y_u, z_u \rangle = \langle m \cdot x_u, m \cdot y_u, m \cdot z_u \rangle$$

   EXAMPLE: $5\vec{a} = 5 \langle 4, -1, 2 \rangle = \langle 20, -5, 10 \rangle$

2. Dot Product: You can multiply two vectors in $\mathbb{R}^3$ to produce a real number using the dot product. To compute a dot product, just multiply x-coordinates together, add what you get when you multiply the y-coordinates together, then add what you get when you multiply the z-coordinates together (Why is this useful? Stay tuned, we will see this again shortly):

   $$\vec{u} \cdot \vec{v} = \langle x_u, y_u, z_u \rangle \cdot \langle x_v, y_v, z_v \rangle = x_u \cdot x_v + y_u \cdot y_v + z_u \cdot z_v$$

   EXAMPLE: $\vec{a} \cdot \vec{b} = \langle 4, -1, 2 \rangle \cdot \langle -2, 3, 5 \rangle = 4 \cdot (-2) + (-1) \cdot 3 + 2 \cdot 5 = -8 - 3 + 10 = -1$

3. Cross Product: You can multiply two vectors in $\mathbb{R}^3$ and get a new vector in $\mathbb{R}^3$. Geometrically, the vector $\vec{u} \times \vec{v}$ constructs a third vector which is perpendicular to both $\vec{u}$ and $\vec{v}$ (Notice how each coordinate is constructed from the other two – the x-coordinate of the product comes from the y's and z's of the inputs, etc., except that the subtraction in the middle coordinate is backwards compared to the other two):

   $$\vec{u} \times \vec{v} = \langle y_u \cdot z_v - z_u \cdot y_v \ , \ z_u \cdot x_v - x_u \cdot z_v \ , \ x_u \cdot y_v - y_u \cdot x_v \rangle$$

Example: $\vec{a} \times \vec{b} = \langle 4, -1, 2 \rangle \times \langle -2, 3, 5 \rangle$
$= \langle (-1) \cdot 5 - 2 \cdot 3 \ , \ 2 \cdot (-2) - 4 \cdot 5 \ , \ 4 \cdot 3 - (-1) \cdot (-2) \rangle$
$= \langle -5 - 6, -4 - 20, 12 - 2 \rangle = \langle -11, -24, 10 \rangle$

[Brief optional aside:] The cross product is very useful geometrically, giving a fairly easy computation to construct perpendicular objects. Algebraically, it is really problematic as a multiplication operation, it does not behave the way you'd want a product to behave. The biggest problem is that the cross product is not *associative*: $(\vec{u} \times \vec{v}) \times \vec{w} \neq \vec{u} \times (\vec{v} \times \vec{w})$. This throws a huge chunk of what we know of algebra out the window. The smaller problem is that it is *anti-commutative*: $\vec{u} \times \vec{v} = -\vec{v} \times \vec{u}$. It was proven in 1958 by Raoul Bott and John Milnor that you can only build associative multiplication operations in 1-dimensional, 2-dimensional and 4-dimensional space. The 2-dimensional version is the *complex numbers* (using $i = \sqrt{-1}$), and the 4-dimensional version is the *quaternions*. I spent a couple years of my life studying John Milnor's work earlier in my life.

Trigonometry: It is helpful to be able to calculate the angle between two vectors in $\mathbb{R}^3$. We will consider the three more basic trigonometric functions (sin, cos, tan), the other three would just be computed using the reciprocal of the appropriate function below.

Two quick thoughts about this:

- We shouldn't get negative angles because the difference between, say, $-45°$ and $45°$ is just a difference of direction, and we don't have a preferred direction here: we are just looking for the angle between $\vec{u}$ and $\vec{v}$, not the angle from $\vec{u}$ to $\vec{v}$ or from $\vec{v}$ to $\vec{u}$.
- We shouldn't get an angle above π radians ($180°$) either, because the angle starting at $\vec{u}$ going all the way around back to $\vec{u}$ is 2π radians ($360°$), so going one way around to $\vec{v}$ gives an angle of π radians ($180°$) or more, and the other way is π radians ($180°$) or less, so we just pick the smaller of the two.

So, to combine these thoughts: The angle between two vectors is always between 0 and π radians (0 and $180°$).

1. Cosine: To compute $\cos(\theta)$ where θ is the angle between $\vec{u}$ and $\vec{v}$, we use the dot product (this is what the dot product is useful for – it allows us to compute the angle between two vectors, this works in any finite number of dimensions; don't get me started on infinite dimensional space):

$$\cos(\theta) = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \, |\vec{v}|}$$

Example: If θ is the angle between $\vec{a}$ and $\vec{b}$, then:

$$\cos(\theta) = \frac{\langle 4, -1, 2 \rangle \cdot \langle -2, 3, 5 \rangle}{|\langle 4, -1, 2 \rangle| \, |\langle -2, 3, 5 \rangle|}$$

$$= \frac{-8-3+10}{\sqrt{16+1+4}\sqrt{4+9+25}} = \frac{-1}{\sqrt{21}\sqrt{38}} = \frac{-1}{\sqrt{798}} \approx -0.0354$$

So, it looks like the angle between $\vec{a}$ and $\vec{b}$ is just a little larger than $\frac{\pi}{2}$ radians (90°), since $\cos(\frac{\pi}{2}) = 0$, and becomes negative only when the angle gets larger than $\frac{\pi}{2}$ radians (90°).

This is an important observation: two vectors are perpendicular exactly when when their dot product is zero, because that is when $\cos(\theta) = 0$. In spaces where the geometry is less obvious, we actually use that as a definition: two objects are defined as perpendicular when their dot product is zero.

2. Sine: There are two ways to compute $\sin(\theta)$ in $\mathbb{R}^3$, take your pick, either the trig identity or the cross product:

$$\sin(\theta) = \sqrt{1-\cos^2(\theta)}$$
$$\sin(\theta) = \frac{|\vec{u}\times\vec{v}|}{|\vec{u}|\,|\vec{v}|}$$

Notice that $\sin(\theta)$ is always positive between 0 and π radians (0 and 180°), so we don't have to worry about the fact that both of those formulas only produce positive numbers.

EXAMPLE: If θ is the angle between $\vec{a}$ and $\vec{b}$, then:

$$\sin(\theta) = \sqrt{1-(-0.0354)^2} \approx 0.9994$$

To double-check, we'll also use the other formula (I'll plug in the values from the examples above):

$$\sin(\theta) = \frac{|\vec{u}\times\vec{v}|}{|\vec{u}|\,|\vec{v}|} = \frac{|\langle -11,-24,10\rangle|}{\sqrt{21}\sqrt{38}} = \frac{\sqrt{121+576+100}}{\sqrt{798}} = \frac{\sqrt{797}}{\sqrt{798}} \approx 0.9994$$

Because the $\sin(\theta)$ is very close to 1, we know that θ is very close to $\frac{\pi}{2}$ radians (90°), but we can't tell whether θ is larger or smaller than that using only $\sin(\theta)$.

3. Tangent: We just use the basic trig identity:

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

EXAMPLE: $\tan(\theta) \approx \dfrac{0.9994}{-0.0354} \approx -28.23$

Again, the $\tan(\theta)$ is a large negative number, so we can see that $x = \theta$ is just beyond the asymptote at $x = -\frac{\pi}{2}$.

4. Inverse Trig: We should also be able to compute the angle itself, not just the values of the trig functions. Notice that $\tan(90°)$ is undefined, so we'd like to avoid that, and, for example, $\sin(30°) = \sin(150°)$, so we'd like to avoid that, so we should choose the arccos() (i.e. $\cos^{-1}()$) function to calculate the angle:

$$\theta = \arccos(\cos(\theta)) = \arccos\left(\frac{\vec{u}\cdot\vec{v}}{|\vec{u}|\,|\vec{v}|}\right)$$

Notice that the statement "the arccos() function" is actually a bit misleading: there are (at least) two versions: one that gives an angle in degrees and one that gives an angle in radians.

EXAMPLE: $\theta = \arccos(\frac{\vec{u}\cdot\vec{v}}{|\vec{u}|\,|\vec{v}|}) = \arccos(-0.0354) = 1.6062$ radians $= 92.03°$